

$$\begin{aligned} \text{[but } 2^3 &= \left[\left(1 + \frac{R}{100} \right)^{15} \right]^3 \text{ from Eq. (i)]} \\ \Rightarrow \left(1 + \frac{R}{100} \right)^n &= 2^3 = \left(1 + \frac{R}{100} \right)^{45} \\ &\text{[on comparing]} \\ \therefore n &= 45 \text{ yr} \end{aligned}$$

Shortcut Method

Since, a sum of money at compound interest doubles itself in 15 yr.

Hence, it will become eight times (2^3 times) in $= 15 \times 3 = 45$ yr

22. (c) Amount due at the end of 1 yr
 $= 4000 \left(1 + \frac{8}{100} \right) = ₹ 4320$

$\therefore$ Amount due after 1 yr
 $= (4320 - 1500) = ₹ 2820$

Amount due at the end of second year
 $= 2820 \left(1 + \frac{8}{100} \right) = ₹ 3045.60$

$\therefore$ Amount due after second year
 $= (3045.60 - 1500) = ₹ 1545.60$

Amount due at the end of third year
 $= 1545.60 \left(1 + \frac{8}{100} \right) = ₹ 1669.25$

Amount due after the payment of third instalment
 $= (1669.25 - 1500) = ₹ 169.25$

23. (a) Let the sum be ₹ x , then
 $x \left(1 + \frac{R}{100} \right)^5 = 2x \Rightarrow \left(1 + \frac{R}{100} \right)^5 = 2$... (i)

The amount after 20 yr,
 $x \left(1 + \frac{R}{100} \right)^{20} = x \left[\left(1 + \frac{R}{100} \right)^5 \right]^4$
 $= 2^4 x = 16x$ [from Eq. (i)]
 $= 16 \times 10000 = ₹ 160000$
 [put $x = ₹ 10000$]

24. (b) Data is insufficient as the rate of interest is not given to calculate further.

25. (d) Let amount be ₹ x and rate of interest is R % annually.

According to the questions,

Amount after 1st yr = ₹ 1200

$$x \left(1 + \frac{R}{100} \right) = 1200 \quad \dots (i)$$

Amount after 3rd yr = ₹ 1587

$$x \left(1 + \frac{R}{100} \right)^3 = ₹ 1587 \dots (ii)$$

On dividing Eq. (ii) by Eq. (i), we get

$$\begin{aligned} \left(1 + \frac{R}{100} \right)^2 &= \frac{1587}{1200} = \frac{529}{400} \\ 1 + \frac{R}{100} &= \frac{23}{20} \Rightarrow \frac{R}{100} = \frac{3}{20} \end{aligned}$$

$\therefore R = 15\%$

Put $R = 15\%$ in Eq. (i),

$$x \left(1 + \frac{15}{100} \right) = ₹ 1200 \Rightarrow x = \frac{1200 \times 100}{115}$$

$\therefore x = ₹ 1043.478$

26. (d) I. Given, $R = 4\%$, $n = 2$ yr and

$$A = ₹ 169, P = ?$$

$$A = P \left(1 + \frac{R}{100} \right)^n \quad \text{[Rule 1. (i)]}$$

$$169 = P \left(1 + \frac{4}{100} \right)^2$$

$$\Rightarrow 169 = P \left(\frac{26}{25} \right)^2$$

$$P = \frac{169 \times 25 \times 25}{26 \times 26} = ₹ 156.25$$

II. Given, $SI = ₹ 120$, $n = 2$ yr

and $CI = ₹ 129$.

$$SI = \frac{P \times R \times T}{100}$$

$$120 = \frac{P \times R \times 2}{100} \Rightarrow PR = ₹ 6000$$

$$\therefore P = \frac{6000}{R} \quad \dots (i)$$

$$CI = P \left[\left(1 + \frac{R}{100} \right)^n - 1 \right] \quad \text{[Rule 1. (ii)]}$$

$$129 = P \left[\left(1 + \frac{R}{100} \right)^2 - 1 \right]$$

$$1290000 = P [(100 + R)^2 - 100^2]$$

$$= P [R^2 + R \times 200]$$

$$= \frac{6000}{R} [R^2 + R \times 200]$$

[from Eq. (i)]

$$1290000 = 6000R + 1200000$$

$$R = \frac{90000}{6000} = 15\%$$

Hence, both statements are correct.

27. (d) $P = ₹ 1600$, $r = 25\%$ and $n = 2$ yr

$$\therefore A = P \left[1 + \frac{R}{100} \right]^n \quad \text{[Rule 1. (i)]}$$

$$= 1600 \left[1 + \frac{25}{100} \right]^2$$

$$= 1600 \times \frac{5}{4} \times \frac{5}{4} = ₹ 2500$$

$\therefore$ Compound interest

$$= ₹ 2500 - ₹ 1600 = ₹ 900$$

28. (b) Refer to example 6.

29. (a) Given, principal = ₹ 15000

Rate, $R = ?$

Difference, $D = ₹ 96$

$$\text{For 2 yr Difference, } D = \frac{PR^2}{(100)^2} \quad \text{[Rule 6 (i)]}$$

$$\Rightarrow 96 = \frac{15000 \times R^2}{100 \times 100}$$

$$\Rightarrow R^2 = 64$$

$$\therefore R = 8\%$$

30. (c) $\therefore$ Let the principal be ₹ x .

$$\text{Then, } SI = ₹ \frac{60x}{100}$$

$$\therefore SI = \frac{\text{Principal} \times \text{Rate} \times \text{Time}}{100}$$

$$\Rightarrow \frac{60x}{100} = \frac{x \times \text{Rate} \times 6}{100} \quad [\because \text{Time} = 6 \text{ yr}]$$

$$60 = \text{Rate} \times 6$$

$$\therefore \text{Rate} = 10\%$$

Again principal = ₹ 12000, Time = 3 yr

$$\text{Amount} = \text{Principal} \left(1 + \frac{\text{Rate}}{100} \right)^{\text{Time}}$$

$$= 12000 \left(1 + \frac{10}{100} \right)^3$$

$$= 12000 \left(\frac{11}{10} \right)^3$$

$$= 12000 \times \frac{11 \times 11 \times 11}{1000}$$

$$= 12 \times 121 \times 11 = ₹ 15972$$

$$\therefore CI = \text{Amount} - \text{Principal}$$

$$= 15972 - 12000 = ₹ 3972$$

31. (b) Refer to example 2.

PROFIT AND LOSS

Generally (2-4) questions have been asked from this chapter. Generally questions are based on basic formulae and tricks explained in this chapter which are helpful to solve the problems more easily and quickly.



Cost Price (CP)

The price, at which an article is bought is called its cost price. All the overhead expenses in the transaction like freight, damage etc., are added to the cost price.

Selling Price (SP)

The price at which an article sold is called its selling price.

Profit ($SP > CP$) When an article is sold at a price more than its cost price, then profit is earned.

- Profit = Selling price – Cost price
- Loss = Cost price – Selling price

Loss ($CP > SP$) When an article is sold at a price lower than its cost price, then loss is incurred.

- Profit % = $\frac{\text{Profit}}{CP} \times 100\% = \frac{(SP - CP)}{CP} \times 100\%$
- Loss % = $\frac{\text{Loss}}{CP} \times 100\% = \frac{(CP - SP)}{CP} \times 100\%$

Note Profit and loss percentage is always calculated as the percentage of CP unless otherwise specified.

List Price/Marked Price

The price of an article which is displayed on the price tag is known as marked price. It is the normal price of the thing/products without a discount. Sometimes the shopkeeper increases or decreases the cost price, then this price is taken as the list price of the article.

Important Rules and Formulae

Rule 1 If there is a profit of $r\%$, then

$$(i) SP = \frac{(100 + r) \times CP}{100} \quad (ii) CP = \frac{100 \times SP}{(100 + r)}$$

EXAMPLE 1. If cost price of a fan is ₹ 720. If there is a profit of $16\frac{2}{3}\%$. Then, selling price is

- a. ₹ 840 b. ₹ 940
c. ₹ 1050 d. None of these

Sol. a. Given, $r = 16\frac{2}{3}\%$ and cost price = ₹ 720

$$\begin{aligned} \therefore \text{Selling price (SP)} &= \frac{CP \times (100 + r)}{100} = \frac{720 \times \left(100 + 16\frac{2}{3}\right)}{100} \\ &= \frac{720 \times \left(100 + \frac{50}{3}\right)}{100} = \frac{720 \times 350}{300} = ₹ 840 \end{aligned}$$

Rule 2 If there is a loss of $r\%$, then

$$(i) SP = \frac{(100 - r)}{100} \times CP \quad (ii) CP = \frac{100 \times SP}{(100 - r)}$$

EXAMPLE 2. When Selling price is ₹ 75 and there is a loss of 12%. Then, cost price is

- a. ₹ 22 b. ₹ 44
c. ₹ 55 d. ₹ 85.22

Sol. d. Given, $r = 12\%$ and selling price = ₹ 75

$$\therefore \text{Cost price} = \frac{100 \times SP}{(100 - r)} = \frac{100 \times 75}{(100 - 12)} = \frac{100 \times 75}{88} = 85.22$$

Rule 3 (i) When there are two successive profits or losses of $x\%$ and $y\%$, then the resultant profit or loss percent is given by

$$\left(\pm x \pm y + \frac{(\pm x)(\pm y)}{100} \right)\%$$

- Note** • Take positive for profit and negative for loss.
• If the sign of result is positive, then there is a total gain.
• If the sign of result is negative, then there is a total loss.

EXAMPLE 3. By selling a watch for ₹ 132 a tradesman got two successive profits of 10% and 20%, then the resultant profit percentage is

- a. 22% b. 30%
c. 32% d. 34%

Sol. c. Given, the successive profits are 10%, 20%.

$$\begin{aligned} \text{So, resultant profit percentage} &= \left(10 + 20 + \frac{20 \times 10}{100} \right)\% \\ &= 32\% \end{aligned}$$

Rule 4 When two different articles are sold at the same selling price, getting gain/loss of $x\%$ on the first and gain/loss of $y\%$ on the second, then the overall % gain or % loss in the transaction is given by

$\left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)} \right]\%$. The above expression represent overall gain or loss accordingly as its sign is positive or negative.

When two different articles are sold at the same selling price getting a gain of $x\%$ on the first and loss of $x\%$ on the second, then the overall % loss in the transaction is given by $\left(\frac{x}{10} \right)^2\%$.

Note That in such questions there is always a loss.

EXAMPLE 4. A trader sells two cycles at ₹ 1188 each and gains 10% on the first and loses 12% on the second. What is the profit or loss percent on the whole?

- a. 1% loss b. 1% gain c. No loss no gain d. 3.2% loss

Sol. d. When there is a profit of $x\%$ and loss of $y\%$, then the resultant

$$\text{Profit/Loss percentage} = \left(x - y - \frac{xy}{100} \right)\%$$

Here, $x = 10\%$ and $y = 12$

$$\therefore \text{Profit/Loss percentage} = \left(10 - 12 - \frac{10 \times 12}{100} \right)\% = -3.2\%$$

EXAMPLE 5. Harish sold two scooters, each for ₹ 24000. If he makes 20% profit on the first and 15% loss on the second. What is his gain or loss percent in the transaction?

- a. loss $\frac{20}{41}\%$ b. gain $\frac{20}{41}\%$ c. loss $\frac{32}{73}\%$ d. gain $\frac{32}{73}\%$

Sol. a. Here, $x = 20$ and $y = -15$

$$\begin{aligned} \therefore \text{Overall gain/loss \%} &= \left[\frac{100(x + y) + 2xy}{(100 + x) + (100 + y)} \right]\% \\ &= \left[\frac{100(20 - 15) + 2 \times 20 \times -15}{(100 + 20) + (100 - 15)} \right]\% \\ &= \frac{-100}{205}\% = \frac{-20}{41}\% \end{aligned}$$

Which represents loss.

EXAMPLE 6. Vikram sold two horses for ₹ 990 each gaining 10% on the one and losing 10% on the other. Find his gain or loss percent.

- a. 1% gain b. 1% loss c. 4% gain d. 4% loss

Sol. b. Here, $x = 10$

$$\therefore \text{Overall loss \%} = \left(\frac{x}{10}\right)^2 \% \\ = \left(\frac{10}{10}\right)^2 \% = 1\%$$

(Rule 5) A person buys two items for ₹ S . One is sold at a loss of $r\%$ and the other at a gain of $R\%$, if each item was sold at the same price, then

$$\text{Cost price of item sold at loss} = \frac{S \times (100 + R)}{(100 - r) + (100 + R)}$$

$$\text{Cost price of item sold at gain} = \frac{S \times (100 - r)}{(100 - r) + (100 + R)}$$

EXAMPLE 7. Sudhir bought two boxes for ₹ 1300. He sold one box at a profit of 20% and the other at a loss of 12%. If the selling price of both boxes is the same, then the cost price of each box is

- a. ₹ 700 and ₹ 500 b. ₹ 750 and ₹ 550
c. ₹ 800 and ₹ 600 d. None of these

Sol. b. Given, cost of two boxes $S = ₹ 1300$, profit on first box $R = 20\%$

and loss on second box $r = 12\%$

$$\text{Cost price of box sold at loss} \\ = \frac{1300 \times (100 + 20)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 120}{88 + 120} = ₹ 750$$

$$\text{Cost price of box sold at gain} \\ = \frac{1300 \times (100 - 12)}{(100 - 12) + (100 + 20)} = \frac{1300 \times 88}{88 + 120} = ₹ 550$$

(Rule 6) If a trader professes to sell his goods at cost price, but uses false weights, then gain %

$$= \frac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$$

EXAMPLE 8. A dishonest dealer sells his goods at cost price, but he uses a weight of 960 gm for the kg weight, then the percentage of gain is

- a. $2\frac{1}{3}\%$ b. $4\frac{1}{6}\%$ c. $6\frac{1}{9}\%$ d. $3\frac{1}{2}\%$

Sol. b. Gain % = $\frac{\text{true weight} - \text{false weight}}{\text{false weight}} \times 100$

$$= \frac{1000 - 960}{960} \times 100 = \frac{40}{960} \times 100 = 4\frac{1}{6}\%$$

(Rule 7) If cost price of ' n ' articles is equal to the selling price of ' m ' articles, then

$$\text{Profit percentage} = \left(\frac{n - m}{m}\right) \times 100\% \quad (n > m)$$

$$\text{and loss percentage} = \left(\frac{m - n}{m}\right) \times 100\% \quad (m > n)$$

EXAMPLE 9. If the cost price of 18 chairs be equal to selling price of 16 chairs. The gain percent is

- a. 12% b. 12.5%
c. 14% d. 15.5%

Sol. b. Given, $n = 18$, $m = 16$ [$\because n > m$]

$$\therefore \text{Profit percentage} = \left(\frac{18 - 16}{16}\right) \times 100 \quad \left[\because \frac{n - m}{m} \times 100\right] \\ = \frac{2}{16} \times 100 = \frac{100}{8} = 12.5\%$$

DISCOUNT

The reduction allowed on the marked price of an article is called as discount. Discount is always reckoned on the marked price.

Selling price = Marked price – Discount

(Rule 8) If discount allowed is $r\%$, then

$$\text{Selling price} = \frac{(100 - r)}{100} \times \text{Marked price}$$

EXAMPLE 10. If the marked price of a fan is ₹ 700 and a discount of 10% is given on it, then what is the selling price of the fan?

- a. ₹ 500 b. ₹ 575
c. ₹ 610 d. ₹ 630

Sol. d. Given, $r = 10\%$

$$\text{Selling price} = \frac{(100 - 10)}{100} \times 700 \\ = \frac{90 \times 700}{100} = ₹ 630$$

(Rule 9) If a shopkeeper marks his items at $x\%$ above the cost price and allows customers a discount of $y\%$ then there is $\left(x - y - \frac{xy}{100}\right)\%$ profit or loss according to positive or negative sign, respectively.

EXAMPLE 11. A dealer marked his goods 20% above the cost price and allows a discount of 10%. Then, his gain percent is

- a. 2% b. 4%
c. 6% d. 8%

Sol. d. Given, $x = 20\%$ and $y = 10\%$

$$\text{Profit/Loss percentage} = \left(x - y - \frac{xy}{100}\right)\% \\ = \left(20 - 10 - \frac{20 \times 10}{100}\right)\% = 8\%$$

Hence, positive sign shows profit i.e. profit of 8%.

Successive Discounts (Discount Series)

Rule 10 Single discount equivalent to three successive discounts $r_1\%$, $r_2\%$ and $r_3\%$

$$= \left[1 - \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right) \left(1 - \frac{r_3}{100} \right) \right] \times 100\%$$

Note This formula also can be apply for more than three successive discounts.

EXAMPLE 12. A single discount which is equivalent to two successive discounts of 20% and 5% is

- a. 20% b. 22% c. 24% d. 25%

Sol. c. Given, $r_1 = 20\%$, $r_2 = 5\%$

$\therefore$ Single discount equivalent to two successive discount 20% and 5%

$$\begin{aligned} &= \left[1 - \left(1 - \frac{20}{100} \right) \left(1 - \frac{5}{100} \right) \right] \times 100\% \\ &= \left(1 - \frac{4}{5} \times \frac{19}{20} \right) \times 100\% = \left(1 - \frac{19}{25} \right) \times 100\% \\ &= \frac{6}{25} \times 100\% = 24\% \end{aligned}$$

Rule 11 If marked price of an item is ₹ x and the successive discount rates are $r_1\%$, $r_2\%$, $r_3\%$ and so on, then selling price of the item

$$SP = x \times \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right) \left(1 - \frac{r_3}{100} \right)$$

EXAMPLE 13. The marked price of a watch is ₹ 1600. The shopkeeper gives successive discount of 10%, $r\%$ to the customer. If the customer pays ₹ 1224 for the watch, then the value of r is

- a. 20% b. 35% c. 37% d. 15%

Sol. d. Given, cost price $x = ₹ 1600$, $r_1 = 10\%$

and $r_2 = r\%$

$$\therefore \text{Selling price} = \text{Cost price} \left(1 - \frac{r_1}{100} \right) \left(1 - \frac{r_2}{100} \right)$$

$$1224 = 1600 \left(1 - \frac{10}{100} \right) \left(1 - \frac{r}{100} \right)$$

$$\Rightarrow 1 - \frac{r}{100} = \frac{1224}{1600 \times 9} \Rightarrow \frac{r}{100} = 1 - \frac{1224}{1440}$$

$$\therefore r = 15\%$$

> PRACTICE EXERCISE

- By selling an article for ₹ 247.50, Sonu get a profit of 12.5%. The cost of the article is
(a) ₹ 220 (b) ₹ 205 (c) ₹ 210 (d) ₹ 200
- If cost price of a fan is ₹ 720 and its SP is ₹ 840. Find the gain percent.
(a) 16% (b) $16\frac{2}{3}\%$ (c) $16\frac{1}{3}\%$ (d) $16\frac{7}{3}\%$
- By selling an article for ₹ 110, a man losses 12%. For how much should he sell it to gain 8%?
(a) ₹ 120 (b) ₹ 125 (c) ₹ 135 (d) ₹ 140
- A man buys 4 tables and 5 chairs for ₹ 1000. If he sells the tables at 10% profit and chairs 20% profit, he earns a profit of ₹ 120. Then, what is the cost of one table?
(a) ₹ 200 (b) ₹ 220 (c) ₹ 240 (d) ₹ 260
- If selling price of 8 articles is equal to the cost price of 10 articles, then per cent gain or loss is
(a) 20% (b) 25% (c) 30% (d) 35%
- A man sells fans at the same price on one he gain 20% and losses 20% on the other. His gain or loss is
(a) 4% loss (b) 4% gain
(c) Neither gain nor loss (d) 1% loss
- Sneha gains 10% on selling a pen. If she sells it at double the price, the profit per cent is
(a) 120% (b) 60% (c) 100% (d) 200%
- On selling an article for ₹ 240, a trader loses 4%. In order to gain 10%, he must sell the article for
(a) ₹ 275 (b) ₹ 280 (c) ₹ 285 (d) ₹ 300
- By selling 8 dozen pencils, a shopkeeper gains the selling price of 1 dozen pencils. What is the gain?
(a) $12\frac{1}{2}\%$ (b) $13\frac{1}{7}\%$ (c) $14\frac{2}{7}\%$ (d) $87\frac{1}{2}\%$
- A man purchased a watch for ₹ 400 and sold it at a gain of 20% of the selling price. The selling price of the watch is
(a) ₹ 300 (b) ₹ 320 (c) ₹ 440 (d) ₹ 500
- A person A sells a table costing ₹ 2000 to a person B and earns a profit of 6%. The person B sells it to another person C at a loss of 5%. At what price did B sell the table?
(a) ₹ 2054 (b) ₹ 2050 (c) ₹ 2024 (d) ₹ 2014
- What price did the seller mark at the printed price of a watch purchased at ₹ 380, so that after giving 5% discount, there is 25% profit?
(a) ₹ 400 (b) ₹ 450 (c) ₹ 500 (d) ₹ 600